9 Markovian Projection Method

9.1 Markovian Projection Lemma:

Strong Convergence of the Itô Differential equation

Lemma 2. The time evolution of the vector of asset prices $\vec{X} \in \mathbb{R}^d$ is given by:

$$dX_i(t) = a_i(t, X(t))dt + b_{ij}(t, X(t))dW_j(t), \ 0 \le t \le T \tag{33}$$

$$\vec{X}(0) = \vec{x}_0 \tag{34}$$

with $\vec{W}$ a k-dimensional Wiener process with independent components.
Consider the (generally) non-Markovian process $S_1 = \vec{P}_1 \cdot \vec{X}$, $\vec{P}_1 \in \mathbb{R}^{d\times 1}$ and the surrogate process $\bar{S}^{x_0} \in \mathbb{R}$, whose dynamics are given by:

$$d\bar{S}^{x_0}(t) = \bar{a}^{x_0}(t, \bar{S}^{x_0}(t))dt + \bar{b}^{x_0}(t, \bar{S}^{x_0}(t))dW(t), \ t \in [0, T] \tag{35}$$

$$\bar{S}^{x_0}(0) = \vec{P}_1 \cdot \vec{x}_0 \tag{36}$$

and the diffusion coefficients $\bar{a}^{x_0}$ and $\bar{b}^{x_0}$ are determined as:

$$\bar{a}^{x_0}(t, s) = \mathbb{E}[\vec{P}_1 \cdot \vec{a}(t, \vec{X}(t))|\vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \tag{37}$$

$$(\bar{b}^{x_0}(t, s))^2 = \ \mathbb{E}[\vec{P}_1^{\,T} b(t, \vec{X}(t))b(t, \vec{X}(t))^T \vec{P}_1|\vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \tag{38}$$

Then, $S_1(t)|_{\vec{X}(0)=\vec{x}_0} = \vec{P}_1 \cdot \vec{X}(t)|_{\vec{X}(0)=\vec{x}_0}$ and $\bar{S}^{x_0}(t)$ have the same conditional distribution for all $t \in [0, T]$.

We note that in general $S_1(t)|_{\vec{X}(0)=\vec{x}_0} = \vec{P}_1 \cdot \vec{X}(t)|_{\vec{X}(0)=\vec{x}_0}$ and $\bar{S}^{x_1}(t)$, where $\vec{x}_1 \neq \vec{x}_0$, do not have the same conditional distribution, although we expect that for $\vec{x}_1$ close to $\vec{x}_0$, the distribution conditional on $\vec{x}_1$ to be well-approximated by the $\vec{x}_0$ coefficients $\bar{a}^{x_0}$ and $\bar{b}^{x_0}$.

Proof:

Let $g : \mathbb{R} \to \mathbb{R}$ and consider the weak approximation error

$$\epsilon_T = \mathbb{E}[g(\bar{S}^{(x_0)}(T)) \mid \bar{S}^{(x_0)}(0) = P_1 x_0] - \mathbb{E}[g(P_1 X(T)) \mid X(0) = x_0]. \tag{39}$$

Let us also define

$$\bar{u}(t, s) = \mathbb{E}[g(\bar{S}^{(x_0)}(T)) \mid \bar{S}^{(x_0)}(t) = s]. \tag{40}$$

Then, we represent the projection weak error as

$$-\epsilon_T = \mathbb{E}[\bar{u}(T, S_1(T)) \mid X(0) = x_0] - \bar{u}(0, S_1(0)) \tag{41}$$

$$= \mathbb{E}\left[\int_0^T (\partial_t \bar{u}(t, S_1(t)) + [\mathcal{G}\bar{u}](t, X(t)))\, dt\right] \tag{42}$$

where the non-Markovian but Itô -integrable process S_1 solves

$$dS_1(t) = P_1 a(t, X(t))dt + P_1 b(t, X(t))dW_t, \tag{43}$$

and we have the generator

$$[\mathcal{G}\bar{u}](t, x) = P_1 a(t, x)\partial_s \bar{u}(t, P_1 x) + \frac{1}{2}(P_1 b b^\top P_1^\top)(t, x)\partial_{ss}\bar{u}(t, P_1 x). \tag{44}$$

Using the fact that $\bar{u}$ solves the projected KBE equation:

$$0 = \partial_t \bar{u} + \bar{\mathcal{G}}\bar{u}, \quad t < T, \tag{45}$$

$$\bar{u}(T, \cdot) = g(\cdot), \tag{46}$$

with coefficients $\bar{a}^{(x_0)}$, $\bar{b}^{(x_0)}$ and

$$\bar{\mathcal{G}}v = \bar{a}^{(x_0)}\partial_s v + \frac{1}{2}(\bar{b}^{(x_0)})^2\partial_{ss}v,$$

we obtain

$$-\epsilon_T = \int_0^T \mathbb{E}_{0,x_0}\left[\left(P_1 a(t, X(t)) - \bar{a}^{(x_0)}(t, P_1X(t))\right)\partial_s\bar{u}(t, P_1X(t))\right]dt \tag{47}$$

$$+\frac{1}{2}\int_0^T \mathbb{E}_{0,x_0}\left[\left((P_1bb^\top P_1^\top)(t, X(t)) - (\bar{b}^{(x_0)})^2(t, P_1X(t))\right)\partial_{ss}\bar{u}(t, P_1X(t))\right]dt. \tag{48}$$

Let us now choose $\bar{a}^{(x_0)}$ and $\bar{b}^{(x_0)}$ such that $\epsilon_T = 0$. Using the tower property and that by conditioning on $P_1X(t)$ then $\bar{u}(t, P_1X(t))$ and $\bar{a}^{(x_0)}(t, P_1X(t))$ become deterministic functions so can be taken out of the inner expectation we obtain:

$$\mathbb{E}_{0,x_0}\left[\left(P_1a(t, X(t)) - \bar{a}^{(x_0)}(t, P_1X(t))\right)\partial_s\bar{u}(t, P_1X(t))\right] \tag{49}$$

$$= \mathbb{E}_{0,x_0}\left[\mathbb{E}\left[\left(P_1a(t, X(t)) - \bar{a}^{(x_0)}(t, P_1X(t))\right)\partial_s\bar{u}(t, P_1X(t)) \mid P_1X(t)\right]\right] \tag{50}$$

$$= \mathbb{E}_{0,x_0}\left[\left(\mathbb{E}[P_1a(t, X(t)) \mid P_1X(t)] - \bar{a}^{(x_0)}(t, P_1X(t))\right)\partial_s\bar{u}(t, P_1X(t))\right]. \tag{51}$$

We note that it is precisely because the generators $\mathcal{G}$ and $\bar{\mathcal{G}}$ both end up depending on $\partial_s\bar{u}_A(t, P_1X)$ which allows the factorisation and ensures that this term can be made to vanish for all bounded and continuous functions g (as well as for all T since $\bar{a}^{(x_0)}$ and $\bar{b}^{(x_0)}$ does not depend on T). A similar derivation applies to the second term. Thus, we conclude that

$$0 = \mathbb{E}\left[g(S_1(T)) - g(\bar{S}^{(x_0)}(T)) \mid X(0) = x_0\right] \tag{52}$$

for all bounded continuous functions g. Therefore, for all fixed T, the random variable $S_1(T) = P_1X(T)$ has the same conditional distribution as $\bar{S}^{(x_0)}(T)$.

9.1.1 Markovian Projection for Pricing European Options under Black-Scholes Model

9.1.2 Estimating the projected volatility under Black-Scholes and verifying Markovian Projection Lemma

Black-Scholes model for d uncorrelated Wiener processes is given by:

$$\begin{aligned} dX_i(t) &= rX_i dt + \sigma X_i dW_i(t) \\ \vec{X}(0) &= \vec{x}_0 \end{aligned} \tag{53}$$

We consider a basket stocks to be equal-weighted such that $\vec{P}\cdot\vec{X} = \frac{1}{d}\sum_i X_i$. Our non-Markovian process $S = \vec{P}\cdot\vec{X}$ is the solution to the SDE:

$$\begin{aligned} dS &= \frac{1}{d}\sum_i rX_i dt + \sigma X_i dW_i(t) \\ S(0) &= \vec{P}\cdot\vec{x}_0 \end{aligned} \tag{54}$$

We now consider the Markovian projection given by:

$$d\bar{S}^{x_0}(t) = \bar{a}^{x_0}(t, \bar{S}^{x_0}(t))dt + \bar{b}^{x_0}(t, \bar{S}^{x_0}(t))dW(t), \ t \in [0, T] \tag{55}$$

$$\bar{S}^{x_0}(0) = \vec{P}_1 \cdot \vec{x_0} \tag{56}$$

Our coefficients $\bar{a}^{x_0}$ and $\bar{b}^{x_0}$ are then given by:

$$\begin{aligned}\bar{a}^{x_0}(t,s) &= \mathbb{E}[\vec{P} \cdot \vec{a}(t, \vec{X}(t)) | \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= \mathbb{E}[r\vec{P} \cdot \vec{X}(t) | \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= rs\end{aligned} \tag{57}$$

$$\begin{aligned}(\bar{b}^{x_0}(t,s))^2 &= \mathbb{E}[\vec{P}^T b(t, \vec{X}(t)) b(t, \vec{X}(t))^T \vec{P} | \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= \mathbb{E}[\frac{1}{d^2} \sum_i \sigma^2 X_i^2 | \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0]\end{aligned} \tag{58}$$

We will need to approximate $\bar{b}^{x_0}$ and to do so we recall that for two random variable X and Y, the conditional expected value $\mathbb{E}[X|Y] = g(Y)$ satisfies the *best mean square approximation*:

$$\mathbb{E}[(X - g(Y))^2] \leq \mathbb{E}[(X - h(Y))^2] \tag{59}$$

for all function h, $\mathbb{E}[h(Y)^2] < \infty$ and so given $V = \{h : \mathbb{E}[h(Y)^2] < \infty\}$:

$$g(Y) = \arg\min_{h \in V} \mathbb{E}[(X - h(Y))^2] \tag{60}$$

This means we can related finding an expected value to an $\mathcal{L}^2$ regression/minimisation problem.

We model the coefficients in the time and space domain such that:

$$(\bar{b}^{x_0})^2(\cdot,\cdot) \approx \arg\min_{h \in V} \int_0^T \mathbb{E}[\Big(\vec{P}^T b(t, \vec{X}(t)) b(t, \vec{X}(t))^T \vec{P} - h(t, \vec{P} \cdot \vec{X})\Big)^2 | \vec{X}(0) = \vec{x}_0] dt \tag{61}$$

Where V will be the set of ansatz functions to model space and time. We can approximate this integral by discretising the time and applying a Monte Carlo estimate for the expected value. Assuming a grid given by $0 = t_0 < t_1 ... < t_N = T$ we obtain that:

$$(\bar{b}^{x_0})^2(\cdot,\cdot) \approx \arg\min_{h \in V} \frac{1}{M} \sum_m \frac{1}{N} \sum_n \Big(\frac{1}{d^2} \sum_i \sigma^2 X_i^m(t_n)^2 - h(t, \vec{P} \cdot \vec{X}^m(t_n))\Big)^2 \tag{62}$$

where $\vec{X}^m(t)$ are independent realisations of $\vec{X}$.

Regression problem:
Assume we have the monomial basis $(\phi_0, \phi_1, ...)$ with $\phi_i(x) = x^i$. To build a basis of polynomials on $\mathbb{R}^2$ we use:

$$\phi_{(i_1,i_2)}(t,x) = t^{i_1} x^{i_2},\ (i_1, i_2) \in \Lambda \tag{63}$$

The ansatz function is based on this basis:

$$h(t,s) = \sum_{(i_1,i_2) \in \Lambda} c_{(i_1,i_2)} \phi_{(i_1,i_2)}(t,s) \tag{64}$$

We now want to derive the normal equation to this end we introduce notation:

$$D_{(m,n),p} = \phi_p\big(t_n,\ P\,X^m(t_n)\big), \quad m = 1,\dots,M,\ n = 0,\dots,N-1,\ p \in \Lambda,$$

$$\big\{D_{k,p}\big\}_{k=1,\dots,MN}^{p\in\Lambda} \ \in\ \mathbb{R}^{MN \times |\Lambda|},$$

$$\psi_k = \frac{1}{d^2} \sum_{i=1}^d \sigma^2 \big(X_i^m(t_n)\big)^2, \quad \psi \in \mathbb{R}^{MN}.$$

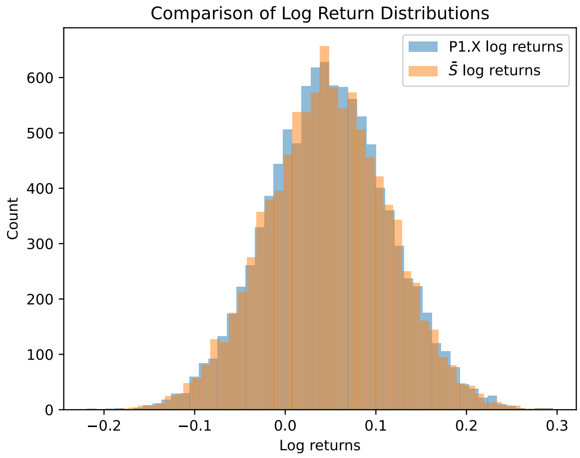

Figure 1: Histogram of the log returns of a basket of 5 stocks as computed by Markov projection $\bar{S}$, where projected volatility was estimated using $\mathcal{L}^2$ regression and from the true process $\vec{P}\cdot\vec{X}$. We observe good agreement between the distribution of the log-returns for both projected and true process.

The optimal coefficients are given by

$$\begin{aligned}\vec{c} \;&=\; \underset{\{c_{i_1,i_2}\}_{(i_1,i_2)\in\Lambda}}{\arg\min}\; \frac{1}{MN}\sum_{m=1}^{M}\sum_{n=0}^{N-1}\left(\frac{1}{d^2}\sum_{i=1}^{d}\sigma^2\,X_i^m(t_n)^2 \;-\; \sum_{(i_1,i_2)\in\Lambda} c_{i_1,i_2}\,\phi_{(i_1,i_2)}\big(t_n, PX^m(t_n)\big)\right)^2 \\ &= \arg\min_{\vec{c}}\,(\psi - D\,\vec{c})^{\mathsf{T}}(\psi - D\,\vec{c})\end{aligned} \tag{65}$$

Equivalently, defining

$$J(\vec{c}) \;=\; \psi^{\mathsf{T}}\psi \;-\; 2\,\vec{c}^{\mathsf{T}} D^{\mathsf{T}}\psi \;+\; \vec{c}^{\mathsf{T}} D^{\mathsf{T}} D\,\vec{c},$$

we set

$$\frac{\partial J(\vec{c})}{\partial \vec{c}} \;=\; -2\,D^{\mathsf{T}}\psi \;+\; 2\,D^{\mathsf{T}} D\,\vec{c} \;=\; 0.$$

We implement this in Python, where we first generate $M_t = 100$ paths of $\vec{X}$ using the forward Euler approximation. After deriving the design matrix D and ψ, we solve the normal equation to obtain the coefficients $\vec{c}$. We then test the Markov projection against the true d-dimensional distribution applying $\vec{P}$ at the final time. We note that the regression becomes more stable with an orthogonal polynomial basis with respect to the density: $\vec{P}\cdot\vec{X}$. Although the exact density is unknown, we can use the empirical density to derive an inner product and an orthogonal basis with respect to this inner product. Our results are shown in Fig.1 for 10^4 Monte Carlo paths.

9.2 Markovian Projection for Pricing American Options

9.2.1 Error bounds

We want to show that:

$$\begin{aligned}u_A(0,\vec{x}) - \bar{u}_A(0,\vec{P}_1\cdot\vec{x}) \;\le\; \int_0^T \mathbb{E}\Big[&\mathcal{H}\Big(t,\vec{X}(t),\vec{P}_1 D\bar{u}_A(t,\vec{P}_1\cdot\vec{X}(t)),\vec{P}_1 D^2\bar{u}_A(t,\vec{P}_1\cdot\vec{X}(t))\vec{P}_1^{\,T}\Big) \\ &- \bar{\mathcal{H}}\big(t,\vec{P}_1\cdot\vec{X}(t),D\bar{u}_A(t,\vec{P}_1\cdot\vec{X}(t)),D^2\bar{u}_A(t,\vec{P}_1\cdot\vec{X}(t))\big)\Big]\,dt\end{aligned} \tag{66}$$

$$\bar{u}_A(0, \vec{P}_1 \cdot \vec{x}) - u_A(0, \vec{x}) \leq \int_0^T \mathbb{E}\Big[- H(t, \bar{S}_1(t)\vec{P}_1, Du_A(t, \bar{S}_1(t)\vec{P}_1), D^2 u_A(t, \bar{S}_1(t)\vec{P}_1)) \\ + \bar{H}(t, \bar{S}_1(t), \vec{P}_1 Du_A(t, \bar{S}_1(t)\vec{P}_1), \vec{P}_1^{\,T} D^2 u_A(t, \bar{S}_1(t)\vec{P}_1)\vec{P}_1)\Big]\, dt \tag{67}$$

This gives us the bounds on the error of using Markovian projection for American options. We explain below why this error arises in the first place.

Define

$$\bar{u}_A(t, s) = \mathbb{E}[g(\bar{S}(T))|\bar{S}(t) = s] \tag{68}$$

with a control given by: $\bar{\alpha}(t, x) = 1_{\bar{u}_A(t, P_1 x) > g(P_1 x)}$.

$$u_A(t, \vec{x}) = \mathbb{E}[g(\vec{P}_1 \cdot \vec{X}(T))|\vec{X}(t) = \vec{x}] \tag{69}$$

with a control given by: $\alpha(t, x) = 1_{u_A(t,x) > g(x)}$.

We note that at $t = T$ (expiry date), $\alpha(T, \vec{X}(T)) = \bar{\alpha}(T, \vec{P}_1 \cdot \vec{X}(T)) = 0$ and $\bar{u}_A(T, \vec{P}_1 \cdot \vec{X}(T)) = u_A(T, \vec{X}(T)) = g(T, \vec{P}_1 \cdot \vec{X}(T))$.

$$\mathbb{E}[\bar{u}_A(T, \vec{P}_1 \cdot \vec{X}(T))|\vec{X}(0) = \vec{x}_0] = \mathbb{E}[g(\vec{P}_1 \cdot \vec{X}(T))|\vec{X}(0) = \vec{x}_0] = u_A(0, \vec{x}_0) \tag{70}$$

$$\begin{aligned} u_A(0, x) - \bar{u}_A(0, \vec{P}_1 \cdot \vec{x}_0) &= \mathbb{E}\left[\bar{u}_A(T, \vec{P}_1 \cdot \vec{X}(T)) \mid X(0) = x_0\right] - \bar{u}_A(0, \vec{P}_1 \cdot \vec{x}_0) \\ &= \mathbb{E}_{0,x_0}\left[\int_0^T dt\ \partial_t \bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)) + [\mathcal{L}\bar{u}_A](t, x)\right] \\ &= \mathbb{E}_{0,x_0}\Bigg[\int_0^T dt\ - \bar{\mathcal{H}}\left(t, \vec{P}_1 \cdot \vec{X}(t), D\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)), D^2\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t))\right) \\ &\qquad + [\mathcal{L}\bar{u}_A](t, x)\Bigg] \\ &\leq \mathbb{E}_{0,x_0}\Bigg[\int_0^T dt\ - \bar{\mathcal{H}}\left(t, \vec{P}_1 \cdot \vec{X}(t), D\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)), D^2\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t))\right) \\ &\qquad + \mathcal{H}\Big(t, \vec{X}(t), \vec{P}_1 D\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)), \vec{P}_1 D^2\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t))\vec{P}_1^{\mathrm{T}}\Big)\Bigg] \end{aligned} \tag{71}$$

The operator $\mathcal{L}v(x, t)$ is given by:

$$\mathcal{L}v(t, x) = \vec{a} \cdot (\vec{P}_1 \partial_x)v(t, x) + \frac{1}{2}Tr[bb^T(\vec{P}_1 \partial^2_{xx} v(t, x)\vec{P}_1^T)] \tag{72}$$

In the proof we have used that $\bar{u}_A$ satisfies the HJB equation given by:

$$-\partial_t \bar{u}_A(t, x) = \big(\bar{\mathcal{H}}\bar{u}_A\big)(t, x),\ (t, x) \in [0, T] \times D, \tag{73}$$

$$\bar{u}_A(t, \cdot) = g(\cdot) \tag{74}$$

and that $[\mathcal{L}\bar{u}_A](t, x) \leq \mathcal{H}$. Analogously, we may prove the other inequality (67) using that the error may be re-written in the form:

$$\bar{u}_A(0, \vec{P}_1 \cdot \vec{x}_0) - u_A(0, \vec{x}) = \mathbb{E}\left[u_A(T, \vec{P}_1 \bar{S}(T)) \mid \bar{S}(0) = \vec{P}_1 \cdot \vec{x}_0\right] - u_A(0, \vec{P}_1 \bar{S}(0)) \tag{75}$$

where we assume that the basket vector $\vec{P}_1$ is normalised.

9.2.2 Source of error in pricing American options using Markovian projection

We now show that if we assume that α is the optimal control for u_A, then the projected coefficients must satisfy:

$$\bar{\alpha}^{x_0}(t,s)\bar{a}^{x_0}(t,s) = \mathbb{E}[\alpha(t,\vec{X}(t))\vec{P_1}\cdot\vec{a}(t,\vec{x}(t))|\vec{P_1}\cdot\vec{X}(t)=s,\vec{X}(0)=\vec{x}_0] \tag{76}$$

$$(\bar{b}^{x_0}(t,s))^2 = \mathbb{E}[\alpha(t,\vec{X}(t))\vec{P_1}^T\vec{b}(t,\vec{X}(t))\vec{b}(t,\vec{X}(t))^T\vec{P_1}|\vec{P_1}\cdot\vec{X}(t)=s,\vec{X}(0)=\vec{x}_0] \tag{77}$$

Proof: Pricing an American Option involves the optimal control of an SDE using a parameter $\alpha\in\{0,1\}$. If we assume that alpha is fixed and known then we obtain a standard SDE for the underlying process is simply given by:

$$dS_1(t) = \alpha(t,X(t))\vec{P_1}\cdot\vec{a}(t,\vec{X}(t))dt + \alpha(t,\vec{X}(t))\vec{P_1}^T b(t,\vec{X}(t))d\vec{W}(t),\ 0\le t\le T \tag{78}$$

$$\vec{S_1}(0) = \vec{P_1}\cdot\vec{x}_0 \tag{79}$$

With $\alpha(t,\vec{X}(t))\in\{0,1\}$. By the Lemma 2 we have that:

$$\bar{\alpha}^{x_0}(t,s)\bar{a}^{x_0}(t,s) = \mathbb{E}[\alpha(t,\vec{X}(t))\vec{P_1}\cdot\vec{a}(t,\vec{x}(t))|\vec{P_1}\cdot\vec{X}(t)=s,\vec{X}(0)=\vec{x}_0] \tag{80}$$

$$(\bar{\alpha}^{x_0}(t,s))^2(\bar{b}^{x_0}(t,s))^2 = \mathbb{E}[\alpha^2(t,\vec{X}(t))\vec{P_1}^T\vec{b}(t,\vec{X}(t))\vec{b}(t,\vec{X}(t))^T\vec{P_1}|\vec{P_1}\cdot\vec{X}(t)=s,\vec{X}(0)=\vec{x}_0] \tag{81}$$

Using that $\alpha^2(t,\vec{X}(t))=\alpha(t,\vec{X}(t))$ and $(\bar{\alpha}^{x_0}(t,s))^2=\bar{\alpha}^{x_0}(t,s)$ we obtain the required result.

From this we can conclude the Markovian projection error arises because one must project the *combination* of the control $\alpha(t,\vec{X}(t))$ and $\vec{P_1}\cdot\vec{a}(t,\vec{X}(t))$ (and likewise for the diffusion coefficient). This means that, in general, simply projecting only $\vec{P_1}\cdot\vec{a}(t,\vec{X}(t))$ and the equivalent diffusion term will not satisfy the above equations and hence not yield the correct coefficients for Markovian projection SDE (correct coefficients are those for which the weak error vanishes). Practically, however, α being unknown at the outset makes it very difficult to satisfy these equations.

9.2.3 Vanishing of error

We want to show that the integrands vanish in regions where the original process and the projected on take the same decision. This entails that Markovian projection errors will arise only in the regions where the control processes make different decisions.

1. Both in stopping region: both Hamiltonians $\mathcal{H}$ and $\bar{\mathcal{H}}$ in both integrands will be 0 so the error is zero.
2. One in stopping region and one in continuation region: then we have the error will, in general, be non-zero since one of the Hamiltonians vanishes but not the other.
3. Both in the continuation region: so $\alpha=\bar{\alpha}=1$.

$$\begin{aligned} I_1 &= \int_0^T dt\,\mathbb{E}_{0,x_0}[\bar{\mathcal{H}}\left(t,\vec{P_1}\cdot\vec{X}(t),D\bar{u}_A(t,\vec{P_1}\cdot\vec{X}(t)),D^2\bar{u}_A(t,\vec{P_1}\cdot\vec{X}(t))\right)\\ &\qquad - \mathcal{H}(t,\vec{X}(t),\vec{P_1}D\bar{u}_A(t,\vec{P_1}\cdot\vec{X}(t)),\vec{P_1}D^2\bar{u}_A(t,\vec{P_1}\cdot\vec{X}(t))\vec{P_1}^{\mathrm{T}})]\\ &= \int_0^T dt\,\mathbb{E}_{0,x_0}[[\mathcal{L}\bar{u}_A](t,x)-[\bar{\mathcal{L}}\bar{u}_A](t,x)] \quad \underbrace{=}_{\text{By our choice of }\bar{a}\text{ and }\bar{b}} \quad 0 \end{aligned} \tag{82}$$

An analogous proof holds for the other integral/ inequality.

9.2.4 Upper and Lower Bounds on the American Option price

Lower Bound: [4] defines two classes of hitting times for the projected value function $\bar{u}_A$:

$$\bar{\tau}^* \equiv \inf\left\{t \in [0,T] :\ \bar{u}_A\left(t, \bar{S}^{(x_0)}(t)\right) = g\left(\bar{S}^{(x_0)}(t)\right)\right\}, \tag{83}$$

$$\bar{\tau}^\dagger \equiv \inf\left\{t \in [0,T] :\ \bar{u}_A\left(t, \vec{P}_1 \cdot \vec{X}(t)\right) = g\left(\vec{P}_1 \cdot \vec{X}(t)\right)\right\}. \tag{84}$$

We also have the hitting time for the value function of the full pricing process u_A given by:

$$\tau^* \equiv \inf\left\{t \in [0,T] :\ u_A\left(t, \vec{P}_1 \cdot \vec{X}(t)\right) = g\left(\vec{P}_1 \cdot \vec{X}(t)\right)\right\}. \tag{85}$$

and the class of hitting times $\mathcal{T}_0$:

$$\mathcal{T}_0 = \{\,\tau : \Omega \to [0,T] \;\Big|\; \{\tau \le t\} \in \mathcal{F}_t \text{ for all } t \in [0,T]\}. \tag{86}$$

$\bar{\tau}^* \in \bar{\mathcal{F}}_t$ but $\bar{\tau}^\dagger \in \mathcal{F}_t$. Since u_A is defined as the stopping time which gives the supremum we see that any $\tau \in \mathcal{T}_0$ gives a lower bound to the American option price. Since $\bar{\tau}^\dagger \in \mathcal{T}_0$ we obtain lower bound of the American option price of:

$$u_A(0, x_0) \ge \mathbb{E}\left[\exp(-r\,\bar{\tau}^\dagger)\, g\left(\vec{P}_1 \cdot \vec{X}(\bar{\tau}^\dagger)\right) \Big| X(0) = x_0\right]. \tag{87}$$

Upper Bound:

Theorem 6. *Rogers's Dual-Representation of the American Option Price [5]: Let $\bar{Z}(t) = e^{-rt} g(\vec{P}_1 \cdot \vec{X}(t))$, let H_0^1 denote the space of all integrable martingales M such that:*

$$\mathbb{E}[\sup_{0 \le t \le T} |M(t)|] < \infty \text{ and } M(0) = 0 \tag{88}$$

Then the price of the American option is given by:

$$u_A(0, \vec{x}_0) = \inf_{M \in H_0^1} \mathbb{E}[\sup_{0<t<T} (\bar{Z}(t) - M(t)) | \vec{X}(0) = \vec{x}_0] \tag{89}$$

Proof:

$$\begin{aligned} Y_0 &= \sup_{0 \le \tau \le T} \mathbb{E}[Z_\tau] \\ &= \sup_{0 \le \tau \le T} \mathbb{E}[Z_\tau - M_\tau] \\ &\le \mathbb{E}[\sup_{0 \le t \le T} (Z_t - M_t)] \end{aligned} \tag{90}$$

Where we have used the since $M(0) = 0$, $\mathbb{E}[M_\tau] = \mathbb{E}[M_0] = 0$ and for any random variable X_t, $\sup_\tau \mathbb{E}[X_\tau] \le \mathbb{E}[\sup_\tau X_\tau]$. Taking the infimum over all $M \in H_0^1$ then proves that Y_0 is bounded by $\mathbb{E}[\sup_{0 \le t \le T} (Z_t - M_t)]$.

We use that Y_t has a Doob-Meyer decomposition:

$$Z_t \le Y_t = Y_0 + M_t^* - A_t^* \tag{91}$$

where M_t^* is a martingale vanishing at 0 and A_t^* is a previsible[1], integrable increasing process also vanishing at 0.

$$\begin{aligned} \inf_{M \in H_0^1} \mathbb{E}[\sup_{0 \le t \le T} (Z_t - M_t)] &\le \mathbb{E}[\sup_{0 \le t \le T} (Z_t - M_t^*)] \\ &\le \mathbb{E}[\sup_{0 \le t \le T} (Y_t - M_t^*)] \\ &\le \mathbb{E}[\sup_{0 \le t \le T} (Y_0 - A_t^*)] \\ &= Y_0 \end{aligned} \tag{92}$$

[1]A process X_t adapted to a filtration $\mathcal{F}_t$ is previsible if, for each time t, X_t is measurable with respect to the σ-algebra generated by all events that happen strictly before t. You can "predict" the value of the process at time t usiong information available strictly before time t.

Where we have used that since A_t^* is an increasing process: $\sup_{0\le t\le T}(Y_0 - A_t^*) = (Y_0 - A_0^*) = Y_0$. The optimising martingale (the martingale reaching the infimum) can be written as:

$$dR^*(t) = e^{-rt}\left((\nabla u_A)^T b\right)(t, X(t))\, dW(t), \quad 0 < t < T \tag{93}$$

$$R^*(0) = 0. \tag{94}$$

Finding an optimising martingale can be very complex, so, [?] propose to approximate the optimising martingale $R(t)$ by a near optimal martingale $R^*(t)$:

$$u_A(0, \mathbf{x}_0) \le \mathbb{E}\left[\sup_{0\le t\le T}\left(\tilde{Z}(q) - R^\star(t)\right) \mid X(0) = \mathbf{x}_0\right], \tag{95}$$

$$\begin{aligned} &dR^\star(t) = \exp(-rt)\,\left((\nabla\overline{u}_A)^T(t, \vec{P}_1\cdot\vec{X}(t))\,\vec{P}_1\cdot\vec{b}(t, X(t))\right)\, dW(t), \quad t\in[0,T],\\ &R^\star(0) = 0. \end{aligned} \tag{96}$$

9.2.5 Least Squares Monte Carlo for American Options

General methodology for LSMC:

1. Use the dynamic programming principle to reduce the American option pricing problem to many subproblems (solving for the American option price backwards from expiry at T, first at $t = T - \Delta t$).
2. Sample points in space at the time $T - \Delta t$.
3. Define a value function via a suitable parametric family of functions. This step is very important as choosing a family of basis functions that poorly approximates will have large impact on the accuracy of the option price. A lot of intuition and insight into the particular problem is required to choose the right parametric family.
4. Perform a regression to obtain the option price at $t = T - \Delta t$.
5. Compare this "holding value" to that of the early exercise price - when the latter exceeds the former we obtain we obtain a stopping time.

Are there cases where LSMC will be obviously better than Markovian projection (MP)?
If you know the right parametric family to use then it is possible but not obvious that LSMC will be better than MP. There are many difficulties with using LSMC including the error propagation

1. The error from the regression estimate of the continuation value is propagated at each step (backwards). The number of samples you need to alleviate the error propagation further grows exponentially and likewise the number of basis functions required also becomes large.
2. Markovian projection is a non-convergent method thus giving us limitation on the error that Markovian projection involves.

9.3 Essential Dimensionality:

In this section, we give important results derived in Let $1 \le n < d$ and $D \subset \mathbb{R}^d$ be a convex set with piecewise smooth boundary.
We call a function $v : D \to \mathbb{R}$ *essentially n-dimensional* if there exist a function $\zeta : \mathbb{R}^n \to \mathbb{R}$ and a matrix $\mathbf{N} \in \mathbb{R}^{n\times d}$ with orthogonal rows such that $v : \mathbb{R}^d \to \mathbb{R}$ is given by

$$v(x) = \zeta(\mathbf{N}x).$$

By extension, we call a differential operator $\mathcal{K}$ essentially n-dimensional if the following backward PDE is well posed:

$$-\partial_t w(t,x) = \mathcal{K}w(t,x), \quad (t,x) \in [0,T) \times D, \qquad w(T,\cdot) = w_T(\cdot), \tag{284}$$

and it has a unique essentially n-dimensional solution for any essentially n-dimensional terminal value w_T. Here we specifically mean that the function ζ may depend on time, that is,

$$w(t,x) = \zeta(t, \mathbf{N}x),$$

but the matrix $\mathbf{N}$ does not.

Lemma 3. Separating essential dimensionality and early exercise If a model has a pricing PDE whose differential operator $\mathcal{L}$ is essentially n-dimensional then an American option price is also essentially n-dimensional if the corresponding payoff is essentially n-dimensional in the same subspace.

Proof: We begin by approximating the price of an American option by a Bermudan option that can only be exercised at M equally-spaced intervals: $t_m = m\frac{T}{M}$. Assume that our model has a pricing PDE whose differential operator $\mathcal{L}$ is essentially n-dimensional and that our payoff function $g = g(Px)$.

Base case: consider time interval $t_{M-1} < t < T$. In this time interval, the option is European. By assumption, our option price in this time interval therefore is essentially n-dimensional so may be expressed as:

$$u(t,x) = \bar{u}(t, Px), \ t_{M-1} < t < T \tag{97}$$

So the value of our option at exercise time t_{M-1} is the maximum of two essentially one-dimensional function:

$$u(t_{M-1}) = \max(u(t^+_{M-1}, x), g(Px)) = \max(\bar{u}(t^+_{M-1}, Px), g(Px)) \tag{98}$$

Now consider an exercise time t_m where we assume that the option price in the interval $t_m < t < t_{m+1}$ is essentially n-dimensional. Then the price of the Bermudan option at a time t_m is given by:

$$u(t^-_m) = \max(u(t^+_m, x), g(Px)) = \max(\bar{u}(t^+_m, Px), g(Px)) = \bar{u}(t_m, Px) \tag{99}$$

From this we can conclude that the price of the Bermudan at all times will be essentially n-dimensional, assuming that it's pricing PDE and the payoff are both essentially n-dimensional on the same subspace.

The American option price is the limit of the Bermudan option price as $M \to \infty$. Hence, we conclude that given our assumption, the American option price is also essentially n-dimensional.

9.3.1 Relationship between dimensional reduction and exactness of Markovian projection for American Option pricing

Lemma 4. If a model has a pricing PDE whose coefficients are essentially n-dimensional and the American option price is also essentially n-dimensional on the same sub-space, then the Markovian projection on that n-dimensional subspace is exact, namely it produces exact values of the American option price.

Proof: First we prove that the Hamiltonians corresponding to u_A and $\bar{u}_A$ coincide pointwise for the case $n = 1$ but this proof may be easily extended to higher n, that is:

$$\mathcal{H}(t, s\vec{P}, v\vec{P}, \vec{P}Q\vec{P}^T) - \bar{\mathcal{H}}(t, s, v, Q) = 0 \tag{100}$$

Where we recall that $\mathcal{H}$:

$$\mathcal{H} = \max_{\alpha \in \{0,1\}} (\alpha \mathcal{L} u_A(t, \vec{x})) \tag{101}$$

$$\mathcal{L}v(t,\vec{x}) = (\vec{a}_i \partial_{x_i})v(t,\vec{x}) + \frac{1}{2}(bb^T)_{ij}(\partial^2_{x_i x_j} v(t,\vec{x}) \tag{102}$$

Assume that the model's pricing PDE has coefficients which are essentially 1-dimensional then:

$$\begin{aligned} \vec{a}(t, \vec{x}) &= \vec{a}'(t, \vec{P} \cdot \vec{x}) \\ b(t, \vec{x}) &= b'(t, \vec{P} \cdot \vec{x}) \end{aligned} \tag{103}$$

Assume also that the American option price is essentially 1-dimensional such that $u_A(t, \vec{x}) = \tilde{u}_A(t, \vec{P} \cdot \vec{x})$. We now consider the coefficients of the Markovian projection:

$$\begin{aligned} \bar{a}^{(x_0)}(t, s) &= \mathbb{E}[\vec{P} \cdot \vec{a}(t, \vec{X}(t)) \mid \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= \mathbb{E}[\vec{P} \cdot \vec{a}'(t, \vec{P} \cdot \vec{X}(t)) \mid \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= \vec{P} \cdot \vec{a}'(t, s) \end{aligned} \tag{104}$$

$$\begin{aligned} (\bar{b}^{(x_0)}(t, s))^2 &= \mathbb{E}[(\vec{P}^T b b^T \vec{P})(t, \vec{X}(t)) \mid \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= \mathbb{E}[(\vec{P}^T b' b'^T \vec{P})(t, \vec{P} \cdot \vec{X}(t)) \mid \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= (\vec{P}^T b' b'^T \vec{P})(t, s) \end{aligned} \tag{105}$$

From this we can see that the SDE for the true process $S = \vec{P} \cdot \vec{X}$ and the SDE for the projected process $\bar{S}$ will coincide exactly. We can now compare the Hamiltonians:

$$\begin{aligned} \mathcal{H}(t, s\vec{P}, v\vec{P}, \vec{P}Q\vec{P}^T) &= \Big(\vec{a}'(t, s) \cdot \vec{P}v + \tfrac{1}{2} Tr\big[(b'\, b'^T\, \vec{P}Q\vec{P}^T\big]\Big)^+, \\ \bar{\mathcal{H}}(t, s, v, H) &= \Big(\bar{a}(t, s)\, v + \tfrac{1}{2}\, \bar{b}^2(t, s)\, Q\Big)^+, \end{aligned} \tag{106}$$

Using the expressions for our coefficients derived above we see that these Hamiltonian's coincide pointwise. Since the two Hamiltonians coincide *pointwise* not just in expectation, the error bounds derived previously imply that:

$$u_A(t, \vec{x}) = \bar{u}_A\big(t, \vec{P} \cdot \vec{x}\big), \quad \forall\, (t, x) \in [0, T] \times D.$$

Hence, we conclude that if the American option price is essentially n-dimensional then the Markovian projection on that n-dimensional subspace is exact, i.e. it produces exact values of the American option price.

9.4 Bachelier Model:

9.4.1 Exactness of Markovian Projection for pricing American Options under the Bachelier Model:

Risk-free measure and no arbitrage give:

$$\vec{a} = r\vec{s} \tag{107}$$

Bachelier Model chooses:

$$b_{ij} = \Sigma_{ij} \tag{108}$$

with Σ a constant matrix. This yields that $\vec{X}(t)$ is Gaussian distributed with $\vec{X}(t) \sim \mathcal{N}(\vec{\mu}, \Gamma(t))$. The projection $S(t) = \vec{P} \cdot \vec{X}(t)$ is again Gaussian distributed with $S(t) \sim \mathcal{N}(\vec{P} \cdot \vec{\mu}, \vec{P}^T\ \Gamma(t)\ \vec{P})$. We can then use the following important property of Gaussian random variables that :

Proposition. If random variables Y and Z follow a *bivariate Gaussian* distribution with means μ_Y, μ_Z, standard deviations σ_Y, σ_Z, and correlation ρ, then

$$\mathbb{E}[\, Y \mid Z = z\,] \;=\; \mu_Y \;+\; \rho\, \frac{\sigma_Y}{\sigma_Z}\, (z - \mu_Z).$$

Thus, the conditional expectation of a Gaussian random variable is *affine* in the conditioning variable Z.

If we take $Z = S(t)$ and $Y = S(t+h)$ then we can show that the conditional mean and variance depend only on Z, yielding that:

$$\mathbb{P}\big(S_1(t+h) \in A \,\big|\, \mathcal{F}_t\big) \;=\; \mathbb{P}\big(S_1(t+h) \in A \,\big|\, S_1(t)\big), \qquad \forall\, A \in \mathcal{B}(\mathbb{R}).$$

This means that the process $S(t)$ is Markovian. Hence, the Markovian projection of the Bachelier Model will give exact results for the true process $S(t)$: the Markovian projection yields an SDE which coincides with the underlying high-dimensional portfolio in law.

9.4.2 Evaluation of coefficients:

The coefficient $\vec{a}(t, \vec{X}(t)) = r\vec{X}(t)$

$$\begin{aligned}\bar{a}^{x_0}(t,s) &= \mathbb{E}[\vec{P}_1 \cdot \vec{a}(t, \vec{X}(t)) | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= \mathbb{E}[r\vec{P}_1 \cdot \vec{X}(t) | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= rs\end{aligned} \tag{109}$$

An additional consequence of having a constant volatility matrix is that the coefficient $\bar{b}$ of the projected process can be evaluated without needing the Laplace approximation to compute high-dimensional integrals:

$$\begin{aligned}(\bar{b}^{x_0}(t,s))^2 &= \mathbb{E}[\vec{P}_1^{\,T} b(t, \vec{X}(t)) b(t, \vec{X}(t))^T \vec{P}_1 | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &\underbrace{=}_{Bachelier} \vec{P}_1^{\,T} \Sigma \Sigma^T \vec{P}_1\end{aligned} \tag{110}$$

10 Exercise Rate Optimisation:

In this section, we follow [6] in giving a brief overview of the current state of the art methods to determine exercise regions for American options. We then give the key results derived in the paper.

$$u(0, s_0) = \sup_{E \subset \mathbb{R}^{1+d}} \mathbb{E}[Y_{\tau_E T} | S_0 = s_0] \tag{111}$$

pick a set E in the time-state plane; stop the first time your path (t, S_t) hits that set; compute the expected payoff; then choose the set giving the largest expectation.

Markov property implies an optimal stopping rule can be chosen to depend only on the current time and current state, not on the full path history. Therefore, any optimal stopping time can be represented as the first entry time into a stopping region E:

$$E = \{(t,s) : v(t,s) = Y_t(s)\} \tag{112}$$

General strategy to find an option price:

1. Choose a parametrisation of the subsets of $\mathbb{R}^{1+d}$
2. Choose an initial guess $E_0 \subset \mathbb{R}^{1+d}$
3. Update to get $E_n \to E_\infty$ and $\Psi(E_n) \to \Psi(E_\infty) = u(s_0)$

Problems with this general strategy:

1. Difficult to come up with a parametrisation all possible exercise regions for path-dependent (exotic) or multi-dimensional options. Analytic guesses possible for simple options but in higher dimensions this because difficult because exercise regions may be disjoint.

2. Unclear to determine the global optimum, given a parametrisation.
3. In general, expectations are replaced with empirical averages. Finite sample paths means that *** is highly noisy and irregular. Sampling noise - when you scan many candidate sets E each $\hat{\psi}(E)$, the largest observed $\hat{\psi}(E)$ will be the one where the noise happened to push the sample average up - selection maximisation bias.

Aim of the paper is to relax the optimisation problem by replacing exercise regions with exercise rates, defining randomised exercise strategies.

10.1 Longstaff-Schwartz Algorithm:

- sensitive to the number of exercise dates (backwards iteration in time):

Let the exercise dates be $t_0 < t_1 < \cdots < t_N = T$. Suppose we have an approximation $v_n(\cdot)$ to the value function $v(t_n, \cdot)$ that can be evaluated at arbitrary state points. Proceed backward in $n = N, N-1, \ldots, 1$ as follows (showing one generic step at t_{n-1}):

1. Generate M samples $s^{(m)} \sim S_{t_{n-1}}$, $m = 1, \ldots, M$.
2. For each $s^{(m)}$, generate K conditional forward samples $s^{(m,k)} \sim S_{t_n} \mid S_{t_{n-1}} = s^{(m)}, \quad k = 1, \ldots, K$.
3. Estimate the *continuation value* from $s^{(m)}$ by the nested Monte Carlo average
$$c^{(m)} := \frac{1}{K} \sum_{k=1}^{K} v_n\big(s^{(m,k)}\big).$$
4. Regress (discrete least squares) onto a chosen approximation space V (e.g. polynomials):
$$p_{n-1} := \arg\min_{p \in V} \sum_{m=1}^{M} \big| p\big(s^{(m)}\big) - c^{(m)} \big|^2.$$
Like in the $\mathcal{L}^2$ regression exercise, you choose a set of basis function $\phi_{i_1,i_2}(t,x)$ and fit coefficients β_{i_1,i_2} and then do least-squares regression to minimise the error between the expected (discounted) continuation.
5. Define the step-back value function by the dynamic programming relation
$$v_{n-1}(s) := \max\{\, g(s),\, p_{n-1}(s)\,\}.$$
For each path you compare the fitted continuation value and immediate exercise payoff.

Iterate backward to obtain $v_0(\cdot)$, an estimate of the option value at time 0.

Main issues with the Longstaff-Schwartz algorithm:

1. Value function has only one continuous derivative at the boundary E_∞. Polynomial regression is poor at (low accuracy in) approximating functions with kinks/low regularity unless the basis is rich (large ansatz space) and many samples necessary.
2. The error from the regression estimate of the continuation value is propagated at each step (backwards). The number of samples you need to alleviate the error propagation further grows exponentially.

10.2 Exercise Rate Optimisation:

11 Multi-Level Monte Carlo (MLMC)

11.1 Introduction:

In this section, we give an overview of the key results on Multi-Level Monte Carlo from [7] and [8]. When do we use Monte Carlo methods?

1. Uncertainty is of high dimension - which can also mean that the number of uncertain input parameters is high.
2. Uncertainty is strongly non-linear.

11.1.1 Standard (simple) Monte Carlo:

Under standard Monte Carlo the estimator of $\mathbb{E}[P]$ is just an average of values of $P(\omega)$ for N independent samples from the probability space $(\Omega, \mathcal{F}, P)$ given by:

$$\mathbb{E}[\hat{P}] = N^{-1} \sum_{i=1}^{N} P(\omega^{(n)}) \tag{113}$$

The variance of the estimate is given by $N^{-1}\mathbb{V}[P]$, giving an rms error of $\mathcal{O}(N^{-\frac{1}{2}})$. An accuracy of ϵ therefore needs $N = \mathcal{O}(\epsilon^2)$ samples.

11.1.2 Control Variates:

11.1.3 General Idea for Multi-Level Monte Carlo:

Suppose we have a sequence $P_0, ..., P_{L-1}$ which approximates P_L with increasing accuracy and cost. Then we can write the expected value of P_L using a telescoping sum:

$$\mathbb{E}[P_L] = \mathbb{E}[P_0] + \sum_{l=1}^{L} \mathbb{E}[P_l - P_{l-1}] \tag{114}$$

This means that we can use the following unbiased estimator of P_L:

$$Y \;=\; N_0^{-1} \sum_{n=1}^{N_0} P_0^{(0,n)} \;+\; \sum_{\ell=1}^{L} \left\{ N_\ell^{-1} \sum_{n=1}^{N_\ell} \left(P_\ell^{(\ell,n)} - P_{\ell-1}^{(\ell,n)} \right) \right\}. \tag{115}$$

The total cost and the total variance are then given by (assuming independent samples):

$$\text{Cost} \;=\; \sum_{\ell=0}^{L} N_\ell \, C_\ell, \qquad \mathbb{V}(Y) \;=\; \sum_{\ell=0}^{L} N_\ell^{-1} \, V_\ell.$$

Assuming a fixed variance, the cost is minimised by choosing N_l to minimise:

$$\sum_{\ell=0}^{L} \left(N_\ell \, C_\ell \;+\; \mu^2 \, N_\ell^{-1} \, V_\ell \right),$$

with μ^2 a Lagrange multiplier. This gives:

$$N_\ell \;=\; \mu \sqrt{\frac{V_\ell}{C_\ell}},$$

Assuming we want an overall variance given by ϵ^2 we require:

$$\mu \;=\; \varepsilon^{-2}\sum_{\ell=0}^{L}\sqrt{V_\ell\, C_\ell},$$

Hence, the optimal total computational cost is given by:

$$C_{\text{total}} \;=\; \varepsilon^{-2}\Big(\sum_{\ell=0}^{L}\sqrt{V_\ell\, C_\ell}\Big)^2. \tag{1.1}$$

What is determined by the particular problem is the relative rate at which the variance decreases with level compared to the increase in the computational cost.

11.1.4 Geometric MLMC:

When the sample space is infinite, such as when we want to simulate SDEs, the output on the level P_ℓ on level ℓ is an approximation of the random variable P, which cannot be simulated.

Assuming that Y is an approximation to $\mathbb{E}[P]$, then the Mean Square Error (MSE) is given by:

$$\text{MSE} \equiv \mathbb{E}\big[(Y-\mathbb{E}[P])^2\big] = \mathbb{V}[Y] + \big(\mathbb{E}[Y]-\mathbb{E}[P]\big)^2.$$

where the first contribution to the error is the statistical sampling error which measures the variance from using a finite number of Monte Carlo samples and the second is the weak error (or bias), which arises from stopping the telescoping sum at level L. We then assume that Y is the multi-level estimator given by:

$$Y = \sum_{\ell=0}^{L} Y_\ell, \qquad Y_\ell = \frac{1}{N_\ell}\sum_{n=1}^{N_\ell}\big(P_\ell^{(\ell,n)} - P_{\ell-1}^{(\ell,n)}\big), \quad P_{-1}\equiv 0.$$

then we obtain that:

$$\mathbb{E}[Y] = \mathbb{E}[P_L], \qquad \mathbb{V}[Y] = \sum_{\ell=0}^{L}\frac{V_\ell}{N_\ell}, \quad V_\ell \equiv \mathbb{V}\big[P_\ell - P_{\ell-1}\big].$$

To ensure that the MSE is less than ϵ^2, it is sufficient to ensure that each contribution to the MSE is less than ϵ^2. [7] then assumes a geometric sequence of levels with the cost increasing exponentially with levels and variance decreasing exponentially with levels.

Giles' MLMC Theorem

Theorem 7. *Let P be a random variable and P_ℓ its level-ℓ approximation. Suppose there exist independent estimators Y_ℓ based on N_ℓ samples, each of cost C_ℓ and variance V_ℓ, and positive constants $\alpha, \beta, \gamma, c_1, c_2, c_3$ with $\alpha \geq \frac{1}{2}\min(\beta,\gamma)$ such that*

Weak Error: $\big|\mathbb{E}[P_\ell - P]\big| \leq c_1\, 2^{-\alpha\ell}$,

$$\mathbb{E}[Y_\ell] = \begin{cases}\mathbb{E}[P_0], & \ell = 0,\\ \mathbb{E}[P_\ell - P_{\ell-1}], & \ell > 0,\end{cases}$$

Variance: $V_\ell \leq c_2\, 2^{-\beta\ell}$,

Cost: $C_\ell \leq c_3\, 2^{\gamma\ell}$.

Then for any $\varepsilon < e^{-1}$ there exist L and $\{N_\ell\}$ such that the estimator

$$Y \;=\; \sum_{\ell=0}^{L} Y_\ell$$

satisfies

$$\mathbb{E}\big[(Y - \mathbb{E}[P])^2\big] \;<\; \varepsilon^2,$$

and its cost C obeys

$$\mathbb{E}[C] \;\leq\; \begin{cases} c_4\,\varepsilon^{-2}, & \beta > \gamma, \\ c_4\,\varepsilon^{-2}(\log\varepsilon)^2, & \beta = \gamma, \\ c_4\,\varepsilon^{-2-(\gamma-\beta)/\alpha}, & \beta < \gamma. \end{cases}$$

Sketch of the proof:
Each of the constraints on the size of the weak error and the statistical error gives a constraint on the number of levels L and the number of samples per level N_ℓ respectively.

1. $\mathbb{E}[Y - P] = \mathbb{E}[\sum_\ell Y_\ell - P] = \mathbb{E}[P_L - P] \leq c_1\, 2^{-2\alpha L} < \frac{1}{\sqrt{2}}\epsilon$. This gives that $L \gtrsim \frac{1}{\alpha}\log_2(\frac{1}{\epsilon})$.
2. We obtain that the optimal N_ℓ is given by: $N_\ell = \frac{2}{\epsilon^2}(\sum_{\ell'=0}^{L}\sqrt{V_{\ell'}\,C_{\ell'}})\sqrt{\frac{V_\ell}{C_\ell}}$.

The optimal computational cost is then, assuming that $N_\ell \propto 2^{-(\beta+\gamma)\frac{\ell}{2}}$ and $N_\ell C_\ell \propto 2^{(-\beta+\gamma)\frac{\ell}{2}}$:

$$C \lesssim \frac{1}{\epsilon^2}(\sum_{\ell'=0}^{L}\sqrt{V_{\ell'}\,C_{\ell'}})^2 \tag{116}$$

We now have three cases depending on the relative magnitudes of β and γ:

1. $\beta > \gamma$: then we have a sequence that converges so the brackets are constant. Hence $C \lesssim c_4\epsilon^{-2}$. In this case, most of the computational cost is from the coarse levels where ℓ is low. To further improve the computational cost we can note that adding the further levels will not significantly add to the computational cost. Hence, the equal error split is sub-optimal. Instead, we may give more error to the statistical error/variance term and so fewer samples N_ℓ will be required.
2. $\beta = \gamma$: then we have $C \lesssim k\ \epsilon^{-2}(L+1)^2 \approx c_4\epsilon^{-2}\log(\epsilon)^2$. In this case, computational cost and contribution to the variance is evenly spread across all the levels.
3. $\beta < \gamma$: then the dominant contribution to the sum is from level L. So we obtain: $C \lesssim k\epsilon^{-2}2^{-(\beta-\gamma)L} \approx \epsilon^{-2-\frac{(-\beta+\gamma)}{\alpha}}$. In this case the dominant computational cost is from the finest levels.

We note that the computational cost for standard Monte Carlo is $\mathcal{O}(\epsilon^{-2-\frac{\gamma}{\alpha}})$.

11.1.5 Multi-Index Monte Carlo (MIMC):

11.2 Multi-Level Monte Carlo for SDEs:

We consider SDE's of the form:

$$\begin{aligned} dS_t &= rS_t dt + \sigma S_t dW_t \\ S_0 &= s_0 \end{aligned} \tag{117}$$

We then want to compute the expected value of $f(S(T))$, where is the function f is assumed for now to be Lipschitz.

A simple Euler-Maruyama discretisation with time step h of the SDE gives:

$$\widehat{S}_{n+1} \;=\; \widehat{S}_n \;+\; a\big(\widehat{S}_n, t_n\big)\, h \;+\; b\big(\widehat{S}_n, t_n\big)\, \Delta W_n.$$

11.2.1 Standard Monte Carlo:

The standard Monte Carlo estimator for $\mathbb{E}[f(S(T))]$ is given by:

$$\widehat{Y} \;=\; \frac{1}{N}\sum_{i=1}^{N} f(\widehat{S}_T^{(i)}).$$

where we implicitly assume a certain step-size h. Assuming that $a(S_t,,t)$ and $b(S_t,t)$ satisfy certain conditions (Bally and Talay 1995, Kloeden and Platen 1992, Talay and Tubaro 1990) then the MSE is given by:

$$MSE \approx c_1 N^{-1} + c_2 h^2 \tag{118}$$

The first term corresponds to the statistical sampling error (variance in the estimator $\hat{Y}$) and the second term the weak error due to error induced by the Euler time discretisation which is $\mathcal{O}(h)$. Setting $MSE = \epsilon^2$, we require $N = \mathcal{O}(\epsilon^{-2})$ and $h = \mathcal{O}(\epsilon)$. This gives a total computational cost $C_{MC} = \mathcal{O}(\epsilon^{-3})$

11.2.2 Multi-Level Monte Carlo for Euler-discretised SDE:

Giles' MLMC Method:

1. We take a geometric sequence of levels such that at level ℓ the uniform time-step is taken to be $h_\ell = h_0 M^{-\ell}$ with M a positive integer typically taken to be 2. h_0 may be taken to be the entire time interval T as long as it remains a good enough control variate for the other finer levels.

2. To couple the coarse and the fine levels we use the same underlying Brownian path and sum the Brownian increments for the fine path time steps to obtain the Brownian increments for the coarse time-steps.

3. We can then use that given a Lipschitz payoff function f for which for any S_1, S_2 there exists a constant K such that $|f(S_1) - f(S_2)| \le K||S_1 - S_2||$ and that the strong error of the Euler discretisation is $\mathbb{E}[||S - \hat{S}||^2] = \mathcal{O}(h)$ that:

$$\begin{aligned} V_\ell \equiv \mathbb{V}[P_\ell - P_{\ell-1}] =& \mathbb{V}[(P_\ell - P) - (P_{\ell-1} - P)] \\ \le& 2(\mathbb{V}[P - P_\ell] + \mathbb{V}[P - P_{\ell-1}]) \\ \le& 2(K^2 \mathbb{E}[||S - \hat{S}_\ell||^2] + K'^2 \mathbb{E}[||S - \hat{S}_{\ell-1}||^2] \end{aligned} \tag{119}$$

   This gives that $V_\ell = \mathcal{O}(h_l)$. Hence, if we take $M = 2$ we have $\alpha = 1, \beta = 1, \gamma = 1$. This means the complexity to reach a rms error of ϵ^2 is, by Theorem [7], $C_{MLMC} = \mathcal{O}(\epsilon^{-2}(\log(\epsilon))^2)$.

This means the MLMC method is near the optimal lower bound of $\mathcal{O}(\epsilon^2)$ for complexity, save logarithmic factors.

11.2.3 Numerical Example: Pricing a European Call Option under GBM

We now price a European Call Option on a single stock, giving pay-off $f(S_T) = e^{-rT}(S_T - K)^+$ whose underlying asset satisfies Geometric Brownian Motion (GBM) following the methodology outlined in [7].

$$dS_t = rS_t + \sigma S_t dW_t \tag{120}$$

Following [7], we choose $r = 0.05, \sigma = 0.2, S_0 = 100, T = 1, K = 100$, allowing us to verify that we do in fact correctly implement MLMC.

This contrasts with the standard Monte Carlo cost which is approximately $\epsilon^{-2} C_L V_0$.

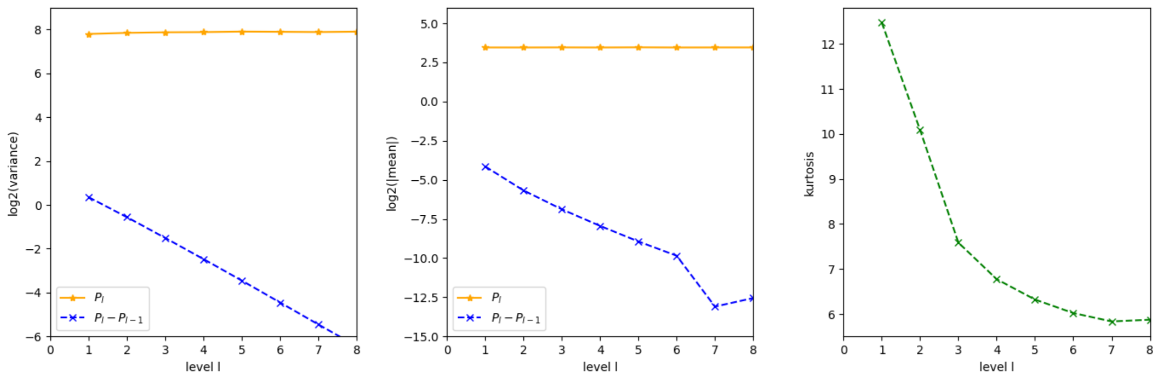

Figure 2: We estimate the variance, absolute mean and kurtosis with level ℓ of an initial sample of 10^5 paths. [7,8] checks the kurtosis given by $\kappa = \frac{\mathbb{E}[X^4]}{(\mathbb{E}[X^2])^2}$ for a random variable X with zero mean. For a large number of sample, the standard deviation of the sample variance of the random variable X is $\sigma_{samplevar} = \sqrt{\frac{\kappa-1}{N}}\mathbb{E}[X^2])$ It is important to check that κ is small to ensure that a reliable estimate of the variance is obtained.

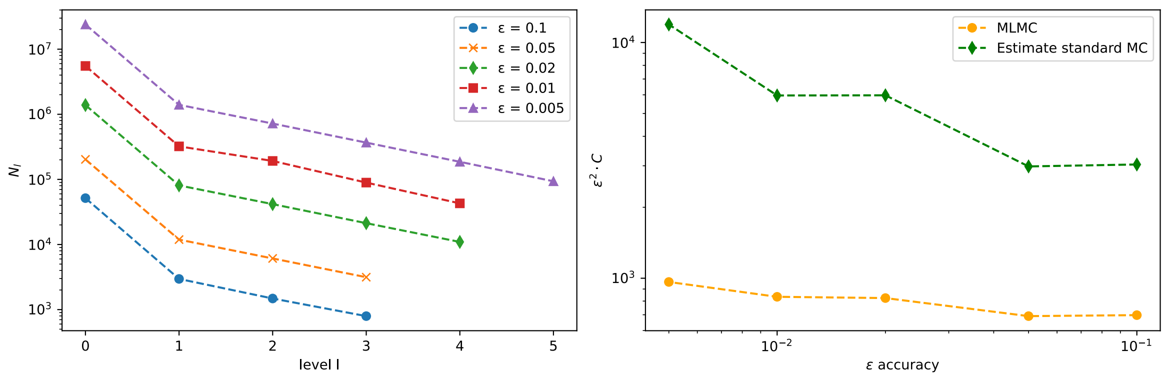

Figure 3: Left hand figure plots number of paths per level for a selected accuracy ϵ. Right hand figure plots $\epsilon^2 * C$ for MLMC and an estimated $\epsilon^2 * C$ of standard Monte Carlo. Given that the theoretical expected cost is for MLMC for our case is $C \sim \mathcal{O}(\epsilon^{-2}(\log(\epsilon))^2)$ we expect that $\epsilon^2 * C$ should be approximately constant as we see in the plot.

11.2.4 Discontinuous Payoffs:

11.3 Multi-Level Weighted Least Squares Approximation [1] [2]

11.3.1 Polynomial (Weighted) Least Squares Approximation:

The goal of polynomial weighted least squares (PWLS) is to approximate a function f, $f \in \mathcal{L}^2_\mu(\Gamma)$, where μ is a probability measure on parameter space Γ, using out finitely many point evaluations of f. The general methodology goes as follows:

1. Choose a finite-dimensional space of polynomials $V \in \mathcal{L}^2_\mu(\Gamma)$.
2. Compute $f(\vec{y_i})$ for random samples $(\vec{y_i})_{i=1}^N$ drawn i.i.d from some distribution ν on Γ
3. Minimise the discrete least squares error:
$$\begin{aligned} \Pi_V f :=& \arg\min_{v\in V} ||f - V||^2_{disc} \\ ||f - V||^2_{disc} :=& \frac{1}{N}\sum_{j=1}^{N} \frac{d\mu}{d\nu}|f(\vec{y_j}) - v(\vec{y_j})|^2 := \langle f - V, f - V\rangle_{disc} \end{aligned} \tag{121}$$
where the $\frac{d\mu}{d\nu} := w$ is the weight function and corrects for the difference between the true distribution μ and the sampling distribution ν.
4. The coefficients $\vec{c}$ of $\Pi_V f$ with respect to any basis $(B_j)_{j=1}^m$ of V are given by the normal equation:
$$\hat{G}\vec{c} = \vec{v} \tag{122}$$
with $\hat{G}_{ij} = \langle B_i, B_j\rangle_{disc}$ and $v_j = \langle f, B_j\rangle_{disc}$ with $i, j \in \{1, ..., m\}$ assuming that $\hat{G}$ is invertible. If $\hat{G}$ is not invertible, then we define $\Pi_V f$ as the one with minimal $\mathcal{L}^2_\mu(\Gamma)$ norm.

Uniqueness of minimisation requires that $N \geq \dim(V) = m$. If $\nu = \mu$, then $N = \mathcal{O}(m^b \log(m))$ ensures a quasi-optimal polynomial fit, where the specific value of $b \geq 1$ depends on μ. For a specific optimal distribution we can obtain the same result with $N = Cm\log(m)$ (Hampton and Doonstan (2015)).

11.3.2 Sampling from the optimal distribution:

We want to sample from the optimal distribution ν^*_V for which $N = \mathcal{O}(m\log(m))$. This occurs for the weight $w^*_V = {\rho^*_V}^{-1}$ which is associated with density:

$$\rho^*_V(y) := \frac{1}{m}\sum_{j=1}^{m}\left|B_j(y)\right|^2.$$

[1] show that the density of the optimal distribution ν^*_V associated with any downward closed[2] polynomial subspace on $[0, 1]^d$ with respect to the Lebesgue measure $d\lambda$ satisfies:

$$C^{-d} < \frac{\mathrm{d}\nu^*_V}{\mathrm{d}\lambda} \leq C^d p^\infty_d,$$

with $0 < C < \infty$ and $p^\infty_d(y)$, the Lebesgue density of the d-dimensional arcsine distribution given by:

$$p^\infty_d(y) := \prod_{j=1}^{d} \frac{1}{\pi\sqrt{y_j\,(1 - y_j)}}.$$

[2] A subspace is downward closed if you have monomial in your subspace and there is another monomial whose degree is smaller in each component, then this monomial is also in your subspace.

The upper bound implies that the number samples for a near-optimal approximation of f, assuming the weight function associated with the arcsine distribution, is given by $C^d m \log(m)$, where C is independent of V (but depends exponentially on d). C^d has been tested to be small even in high dimensions, (≈ 4) for $d = 10$. **As a consequence the upper bound allows for efficient sampling in many cases (without the need for MCMC methods or acceptance/rejection sampling) from the arcsine distribution rather than the optimal distribution for all polynomial subspaces. It also gives a simple weight function independent of V, which is important for adaptive algorithms.**

One way to sample from the univariate arcsine distribution:

1. Generate a uniformly distributed random variable $X \sim U[-\frac{\pi}{2}, \frac{\pi}{2}]$.
2. Transform X via: $Y = \frac{1}{2}(1 + \sin(X))$ then Y samples the univariate arcsine distribution.

11.3.3 Single-Level Approach to Inexact Solutions:

The single-level approach applies the weighted least squares approximation to a single f_n approximating some function f. The problem with this approach is that a good approximation requires a large subspace V and fine time steps leading to many, expensive samples.

It may be shown that if the following two assumptions hold:

1. There are $\beta, \gamma > 0$ such that $||f - f_h||_{\mathcal{L}^2_\mu(\Gamma)} \lesssim h^\beta$ and each evaluation of f_h requires work $h^{-\gamma}, \forall\, h > 0$.
2. There is an $\alpha > 0$ and a sequence of subspaces $(V_n)_{n=1}^{\infty}$ such that $\inf\limits_{v \in V_n} ||f_h - v||_{\mathcal{L}^\infty(\Gamma)} \lesssim (\dim(V_n))^{-\alpha}$ for all $h > 0, n \in \mathbb{N}$

then an accuracy of $\epsilon > 0$ requires $h \approx \epsilon^{-\frac{\gamma}{\beta}}$ and $\dim(V_n) \approx \epsilon^{-\frac{1}{\alpha}}$. The total work therefore is given by:

$$\begin{aligned} &\approx h^{-\gamma} \dim(V_n) \log(\dim(V_n)) \\ &\approx \epsilon^{-\frac{\gamma}{\beta}} \epsilon^{-\frac{1}{\alpha}} |\log(\epsilon)| \end{aligned} \tag{123}$$

Where $\epsilon^{-\frac{\gamma}{\beta}}$ is the contribution to the work from PDE solutions and $\epsilon^{-\frac{1}{\alpha}}$ the contribution from the complexity of polynomial spaces.

11.4 Multi-Level Approach to Inexact Solutions

11.4.1 Core Assumptions:

[1,2] show how the Multi-Level methodology can be applied to improve the computational work required for a weighted least squares approximation.

Authors consider the approximation of a function $f_\infty : \Gamma \subset \mathbb{R}^d \to \mathbb{R}$, $d \in \mathbb{N} \cup \{\infty\}$ in a normed vector space $(F, \|\cdot\|_F) \hookrightarrow (L^2_\mu(\Gamma), \|\cdot\|_{L^2_\mu(\Gamma)})$ of continuous functions on Γ, under the following assumptions:

A1 (Convergence of approximations). There exist functions $f_n \in F$, $n \geq 1$ such that

$$\|f_\infty - f_n\|_F \lesssim n^{-\beta_s}$$

$$\|f_\infty - f_n\|_{L^2_\mu(\Gamma)} \lesssim n^{-\beta_w}$$

for some $\beta_s > 0$ and $\beta_w \geq \beta_s$.

A2(p) (Polynomial approximability). There exist downward closed spaces of polynomials V_m, $m \geq 1$ on Γ such that

$$\dim V_m \lesssim m^\sigma, \quad e_{m,p}(F) \lesssim m^{-\alpha}$$

for some $\sigma > 0$, $\alpha > 0$, and $p = 2$ or $p = \infty$, where

$$e_{m,2}(F) := \sup_{f\in F} \frac{e_{V_m, \|\cdot\|_{L^2_\mu(\Gamma)}}(f)}{\|f\|_F}, \quad e_{m,\infty}(F) := \sup_{f\in F} \frac{e_{V_m,\|\cdot\|_\infty}(f)}{\|f\|_F}.$$

We can understand this assumption as follows: there is a sensible polynomial hierarchy V_m whose size grows like m^σ and for which the best weighted L^p approximation error of any $f \in F$ decays like $m^{-\alpha}$ once normalised by $||f||_F$.

A3 (Sample work). The work required for a single evaluation of f_n satisfies

$$\text{Work}(f_n) \lesssim n^\gamma$$

for some $\gamma > 0$.

11.4.2 Multi-Level Estimator:

We let $h_l \to 0$ (which authors take to be the finite element of diameter l^{-1} but could in principle also be time step of an SDE) in the limit $l \to \infty$ and consider a sequence of functions $f_{h_0}, ..., f_{h_l}$ which approximate in some suitable sense a function f. We then write f as a telescoping sum:

$$f = f_{h_0} + \sum_{l=1}^{\infty} (f_{h_l} - f_{h_{l-1}}) \tag{124}$$

As before, we cut-off this sum at some suitable L. Note that the coarsest approximation of f, f_{h_0}, is taken to be of the same order as f and it is cheap to evaluate many samples. Further, $(f_{h_l} - f_{h_{l-1}}) \to 0$ in the limit $l \to \infty$ and samples are (increasingly) expensive to evaluate.

Choosing that $(h_l)_{l=1}^\infty$ is exponentially decreasing and our sequence of subspaces $(V_l)_{l=1}^\infty$ to be exponentially increasing we can write:

$$\mathcal{S}_L(f) := \Pi_{V_L} f_0 + \sum_{l=1}^{L} \Pi_{V_{L-l}} (f_l - f_{l-1}) \tag{125}$$

where the operator $\Pi_{V_l} : F \to V_l$, for $0 \le l \le L$, is the random weighted least square approximation which projects onto the subspace V_l . Because the projection at each level differs this sum does not telescope in the same way as (124), nor is it required to do so (all that is required is that the polynomial approximability assumption is fulfilled). This entails we get error contributions from each level l. However, in the asymptotic limit $l \to \infty$ V_l goes to infinity and $f_l - f_{l-1} \to 0$. This entails that our error goes to zero exponentially fast.

11.4.3 Multi-Level Convergence Analysis

Multi-Level Convergence Analysis

If $\|f - f_h\|_F \lesssim h^\beta$ and each evaluation of f_h requires work $h^{-\gamma}$, $\forall h > 0$ for some space F for which $\inf_{v\in V_l} \|g - v\|_{L^\infty(\Gamma)} \lesssim \|g\|_F (\dim V_l)^{-\alpha}$, $\forall g \in F$, $l \in \mathbb{N}$, then the multilevel polynomial approximation (with optimal sampling) requires the work

$$W \simeq \varepsilon^{-\max\{\gamma/\beta,\, 1/\alpha\}} |\log \varepsilon|^r,$$

(with some $r = r(\alpha, \beta, \gamma) > 0$ whose value is known) to achieve

- *Convergence in probability:*

$$\|S_L(f) - f\|_{L^2_\mu(\Gamma)} \le \varepsilon \tag{126}$$

  with probability $> 1 - \varepsilon^{|\log \varepsilon|}$, $\quad 0 < \varepsilon \ll 1$.

- *Mean square convergence* (if L^∞ bound is available):

$$\mathbb{E}\big\|\tilde{S}_L(f) - f\big\|^2_{L^2_\mu(\Gamma)} \leq \varepsilon^2.$$

Here we follow [2] in considering the particular case where $\alpha, \gamma, \beta = 1$ and so we aim to show that $W \simeq \epsilon^{-1}|\log(\epsilon)|^r$, for some r depending on α, β, γ

Proof. First we derive an error bound. For any $L > 0$ we have

$$\begin{aligned}
\big\|S_L(f) - f_{h_L}\big\|_{L^2_\mu} &= \big\|\Pi_{V_L} f_0 + \sum_{l=1}^{L} \Pi_{V_{L-l}}(f_l - f_{l-1}) - \Big(f_0 + \sum_{l=1}^{L}(f_l - f_{l-1})\Big)\big\|_{L^2_\mu} \\
&= \big\|(\Pi_{V_L} - \mathrm{Id})\, f_0 + \sum_{l=1}^{L}(\Pi_{V_{L-l}} - \mathrm{Id})\,(f_l - f_{l-1})\big\|_{L^2_\mu} \\
&\leq \underbrace{\|\Pi_{V_L} - \mathrm{Id}\|_{F\to L^2}}_{\text{operator norm}} \|f_0\|_F + \sum_{l=1}^{L} \|\Pi_{V_{L-l}} - \mathrm{Id}\|_{F\to L^2} \|f_l - f_{l-1}\|_F \\
&\leq \|\Pi_{V_L} - \mathrm{Id}\|_{F\to L^2} \|f_0\|_F + \sum_{l=1}^{L} \|\Pi_{V_{L-l}} - \mathrm{Id}\|_{F\to L^2} \|f_l - f\|_F.
\end{aligned}$$

Assuming $\dim V_l = e^l$ and $h_l = e^{-l}$, one has $\|\Pi_{V_l} - \mathrm{Id}\| \approx e^{-l} \approx \|f_l - f\|_F$. Hence

$$\begin{aligned}
\|S_L(f) - f\| \;&\leq\; \|S_L(f) - f_{h_L}\| + \|f_{h_L} - f\| \\
&\lesssim\; e^{-L} + \sum_{l=1}^{L} e^{-(L-l)}\, e^{-l} + e^{-L} \;=\; e^{-L}\,(L+2).
\end{aligned}$$

We note that in the asymptotic limit $L \to \infty$ the error goes to 0. Next we derive the computational complexity of the Multi-Level Polynomial:

$$\begin{aligned}
\mathrm{Work}\big(S_L(f)\big) &= \mathrm{Work}\big(\Pi_{V_L} f_0\big) + \sum_{l=1}^{L} \mathrm{Work}\big(\Pi_{V_{L-l}}(f_l - f_{l-1})\big) \\
&\approx \dim V_L\, \log\!\big(\dim V_L\big)\, \mathrm{Work}(f_0) + \sum_{l=1}^{L} \dim V_{L-l}\, \log\!\big(\dim V_{L-l}\big)\, \mathrm{Work}(f_l - f_{l-1}) \\
&\lesssim e^L\, L \;+\; \sum_{l=1}^{L} e^{\,L-l}\,(L-l)\, e^l \\
&\approx e^L\,(L+1)^2.
\end{aligned}$$

Denoting the right-hand side of the error bound $e^{-L}(L+2) = \epsilon$, we may rephrase the work bound as

$$\mathrm{Work}\big(S_L(f)\big) \;\lesssim\; \varepsilon^{-1}\,\big|\log\varepsilon\big|^3 \;=\; \varepsilon^{-\max\{\gamma/\beta,\ 1/\alpha\}}\,\big|\log\varepsilon\big|^3.$$

$\square$

11.5 Application of Multi-Level Least Squares Polynomial Regression Method to SDEs:

11.5.1 Original Example: Random PDEs

Let $f(\mathbf{y}) = Q(u_{\mathbf{y}})$, where $u_{\mathbf{y}}$ is the solution to a PDE with random parameters $\mathbf{y}$. For example:

$$\begin{aligned} -\nabla\cdot\big(a(x,\boldsymbol{y})\,\nabla u(x,\boldsymbol{y})\big) &= g(x) && \text{in } U \subset \mathbb{R}^D, \\ u(x,\boldsymbol{y}) &= 0 && \text{on } \partial U, \end{aligned} \tag{127}$$

with $a : U \times \Gamma \to \mathbb{R}$ and $\Gamma := [0,1]^d$.

We consider the case where $f(\mathbf{y})$ cannot be evaluated exactly because we do not have access to an analytic solution of the PDE. In this case, we solve the PDE with finite difference or finite element solver giving approximations of f: $f_n(\mathbf{y}) = Q(u_{\mathbf{y},n})$. bo The general methodology to apply multilevel least squares regression then proceeds as follows:

1. Solve the PDE with with the coarsest resolution h_0 for a large number $|\Gamma_L|$ of randomly chosen values of the parameters $\mathbf{y}$.
2. Extend these results to the entire parameter domain by means of a weighted least squares approximation in the large polynomial subspace V_L.
3. Solve the PDE with a finer resolution h_1 and for the coarsest resolution h_0 for a smaller number $|\Gamma_{L-1}|$ of parameters $\mathbf{y}$.
4. Project the difference between coarse and fine resolution solutions onto the smaller polynomial subspace V_{L-1}.
5. Repeat for all levels l.
6. Sum all of the projections to obtain the final multi-level estimator for the function f.

Of critical importance is that the samples y used to compute f_l and f_{l-1} at a given level are the same. This guarantees that the sum (124) telescopes and our assumptions for the multi-level approximation are upheld.

11.5.2 Application to SDEs: Initial issues

We want to estimate:

$$(\bar{b}^{x_0}(t,s))^2 = \mathbb{E}[\vec{P_1}^T \vec{b}(t,\vec{X}(t))\vec{b}(t,\vec{X}(t))^T \vec{P_1} | \vec{P_1}\cdot\vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \tag{128}$$

Where in the Black-Scholes model we have:

$$\vec{b}(t,X(t))_{ij} = X_i\Sigma_{ij} \tag{129}$$

Using the best mean square approximation we can relate finding an expectation value to an $\mathcal{L}^2$ regression problem:

$$\begin{aligned} \mathbb{E}[X|Y] &= g(Y) \\ g(Y) &= \arg\min_{h\in V} \mathbb{E}[(X-h(Y))^2],\ V = \{h : \mathbb{E}[h(Y)^2] < \infty\} \end{aligned} \tag{130}$$

Single-Level:

Model coefficients in time and space domain:

$$(\bar{b}^{x_0})^2(\cdot,\cdot) \approx \arg\min_{h\in V} \int_0^T \mathbb{E}[\Big(\vec{P}^T b(t,\vec{X}(t))b(t,\vec{X}(t))^T\vec{P} - h(t,\vec{P}\cdot\vec{X}(t)\Big)^2 |\vec{X}(0)=\vec{x}_0]dt \tag{131}$$

And then apply a Monte Carlo estimator:

$$(\bar{b}^{x_0})^2(\cdot,\cdot) \approx \arg\min_{h\in V} \frac{1}{M}\sum_m \frac{1}{N}\sum_n \Big(\vec{P}^T b(t_n,\vec{X}^m(t_n))b(t_n,\vec{X}^m(t_n))^T\vec{P} - h(t_n,\vec{P}\cdot\vec{X}^m(t_n))\Big)^2 \tag{132}$$

where $\vec{X}^m(t)$ are independent realisations of $\vec{X}$. For ease of notation, define $g := \vec{P}_1^T\vec{b}(t,\vec{X}(t))\vec{b}(t,\vec{X}(t))^T\vec{P}_1$.

$$\mathcal{S}_L(f) := \Pi_{V_L} g_0(\vec{X}^0) + \sum_{l=1}^L \Pi_{V_{L-l}}(g(\vec{X}^l) - g(\vec{X}^{l-1})) \tag{133}$$

where

$$\Pi_{V_{L-l}}(g(\vec{X}^l) - g(\vec{X}^{l-1})) := \arg\min_{v\in V_{L-l}} ||(g(\vec{X}^l) - g(\vec{X}^{l-1})) - v(\ )|| \tag{134}$$

where we have purposefully left the argument(s) of v blank. The key difference between the case of random PDEs and this SDE case is that the data is a function of **two** different parameters $\vec{X}^l$ and $\vec{X}^{l-1}$.

This gives us (at least) two options:

1. We **assume** that there exists a map $\theta_{\Delta t} : \vec{X}^l \to \vec{X}^{l-1}$ such that in the limit $\Delta t \to 0$, $\vec{X}^{l-1} = \vec{X}^l$. Then we can write:

$$\Pi_{V_{L-l}}(g(\vec{X}^l) - g(\vec{X}^{l-1})) := \arg\min_{v\in V_{L-l}} ||(g(\vec{X}^l) - g(\theta_{\Delta t}(\vec{X}^l))) - v(\vec{X}^l)|| \tag{135}$$

   This map will be dependent on the particular path in question: $\theta_{\Delta t}(\vec{X}^l,\omega)$. The question is whether such a map exists and whether its properties allow for a telescoping sum to be written.

2. We allow that v is a function of two variables $v = v(\vec{X}^{l-1},\vec{X}^l)$. Then we can write:

$$\Pi_{V_{L-l}}(g(\vec{X}^l) - g(\vec{X}^{l-1})) := \arg\min_{v\in V_{L-l}} ||(g(\vec{X}^l) - g(\vec{X}^{l-1})) - v(\vec{X}^{l-1},\vec{X}^l)|| \tag{136}$$

   Since the function to be estimate $\bar{b}^2(t,s)$ is a function of only one variable and assuming that X_l and X_{l-1} are close, we stipulate that $\bar{b}^2(t,s) = v(s,s)$. That is, we evaluate v along the diagonal. Important considerations will be i) determining an appropriate ansatz for v and ii) limiting the computational cost of doubling the dimension. We note that on the coarsest level, the assumption that X_l and X_{l-1} are close will be most strained since this is when the coarse and fine paths are furthest apart.

11.5.3 Application to SDEs: Option 1

We need:

1. Sequence of exponentially increasing polynomial subspaces $(V_l)_{l=0}^\infty$: We chose subspaces of (total degree) Legendre polynomials which are orthogonal on the interval $[-1,1]$ with weight function $w = 1$.

2. Exponentially decreasing time steps $h_0, h_1, ..., h_l, ...$: We chose $h_l = h_0 \cdot 2^{-l}$.

We simulate M_l fine paths $X^f(t^l)$ and construct the coarse paths $X^c(t^{l-1})$ by summing the Brownian increments of the fine paths. We then have a choice:

1. Interpolate the coarse path on the missing fine time steps using a Brownian Bridge.

2. Regress the difference only on the coarse time intervals.

We assumed that the option 2 is sufficient. Further, we chose that $h = h(t_n, \vec{P} \cdot \vec{X}_f^m(t_n^c))$, so h was a function of the fine paths. For $l > 0$, the Multilevel Monte Carlo least squares estimator for the projected diffusion coefficient can then be written as follows:

$$\begin{aligned}(\bar{b}_l^{x_0})^2(\cdot,\cdot) - (\bar{b}_{l-1}^{x_0})^2(\cdot,\cdot) \approx \arg\min_{h \in V} \frac{1}{M_l} \sum_{m=1}^{M_l} \frac{1}{N_{l-1}} \sum_{n=1}^{N_{l-1}} \Big(&[\vec{P}^T b(t_n, \vec{X}_l^m(t_n^{l-1})) b(t_n, \vec{X}_l^m(t_n^{l-1}))^T \vec{P} \\ &- \vec{P}^T b(t_n, \vec{X}_{l-1}^m(t_n^{l-1})) b(t_n, \vec{X}_{l-1}^m(t_n^{l-1}))^T \vec{P}] - h(t_n^{l-1}, \vec{P} \cdot \vec{X}_l^m(t_n^{l-1})) \Big)^2\end{aligned} \tag{137}$$

11.5.4 Application to SDEs: Optimal Transport Map

In some applications, one seeks to construct a *measure-preserving transformation* from a one-dimensional continuous random variable X to another random variable Y, such that the push-forward of the law of X under the map coincides with the law of Y. More precisely, we look for a measurable map

$$T : \mathbb{R} \to \mathbb{R}, \quad \text{such that} \quad T_{\#}\mu_X = \mu_Y,$$

where μ_X and μ_Y denote the probability measures of X and Y, and $T_{\#}\mu_X$ denotes the push-forward measure. Among the infinitely many maps that may satisfy this property, *optimal transport* (OT) theory selects the one that minimizes a given transportation cost functional. A common choice in the quadratic case is

$$T^* = \arg \min_{T_{\#}\mu_X = \mu_Y} \mathbb{E}\left[\, |T(X) - X|^2 \,\right],$$

which, under mild regularity assumptions, admits a unique monotone non-decreasing solution in the one-dimensional setting [9–11].

In practice, when only empirical samples $\{(x_i, y_i)\}_{i=1}^N$ are available, the OT map can be approximated via *discrete optimal transport algorithms*. These include the classical *Hungarian algorithm* for the exact discrete assignment problem and modern *entropic regularization* techniques [12], which lead to the *Sinkhorn algorithm*—a scalable method widely used in machine learning and data-driven applications. For the one-dimensional case, the OT map simplifies to a *quantile map*

$$T^*(x) = F_Y^{-1}(F_X(x)),$$

where F_X and F_Y are the cumulative distribution functions (CDFs) of X and Y, respectively. This formula provides an explicit and computationally trivial way to transport samples from X to the law of Y.